

THE OHIO STATE UNIVERSITY

COLLEGE OF ENGINEERING

ML Decoders for Quantum Error Correction

Team 22

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Meet Team 22



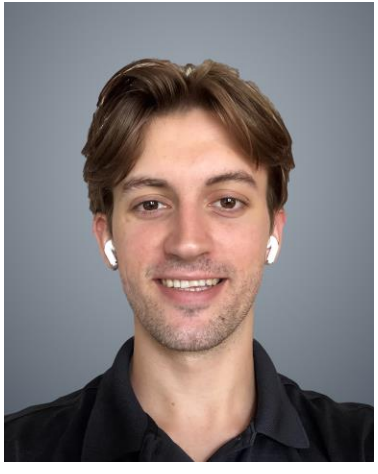
Ali Moallin

Data Pipeline Engineer



Carson Brice

QEC & ML Theory Lead



Daniel Kowalski

Machine Learning Engineer

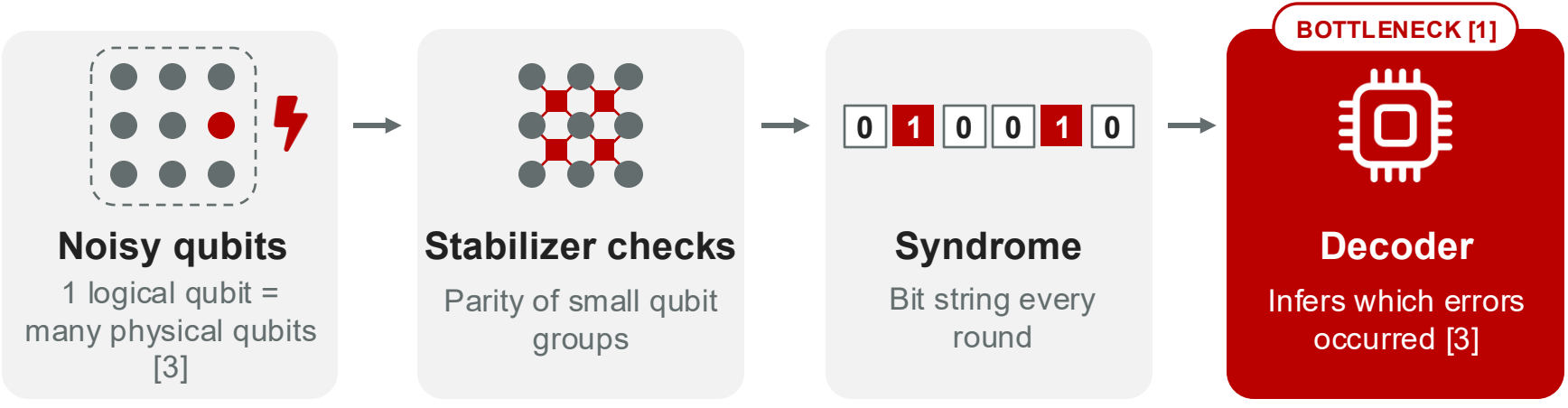


Nate Herman

Data Scientist

Background: Error Decoding

Decoding is a primary bottleneck for fault-tolerant quantum computers [1]



Fast: ~1 µs per round [6], [7]

Accurate: avoid logical errors [3]

Problem Definition

MWPM
Standard decoder [4], [5]

- Hand-designed
- Simplified noise model

vs

ML decoders
Learn from data [6]

- ✓ Fewer logical errors [6]
- ✗ **Too slow for real time [6], [7]**

Two ML decoders that outperform MWPM within size and latency limits

Problem Scope

1. Generate data
Labeled syndromes from Stim [8]

2. MWPM baseline
Accuracy reference

3. Two ML decoders
Feed-forward / CNN
Recurrent / Transformer

4. Compare
Accuracy and size vs. MWPM

Train: Stim simulations [8]

Test: real quantum hardware datasets

Out of scope (stretch goal): FPGA implementation at ~1 µs [6], [7]

Constraints & Assumptions

Constraints

- **No quantum computer** — BIGGEST
- Fixed course timeline
- No high-power GPUs
- Python only
- Required model types

Assumptions

- **Public datasets are labeled and similarly formatted**
- **Stim's simulated noise ≈ real hardware**
  Based on AlphaQubit [6]

Stakeholders



Dr. Kim

Project sponsor

Proof-of-concept ML decoders



Team 22

Capstone team

Graduation and new skills



Future students

Dr. Kim & Dr. Gupta

Reusable code and data



Dr. Z

Course instructor

On-time, quality work



QEC researchers

Academia & industry

Reproducible benchmarks

Needs Assessment

RANK	NEED	STAKEHOLDERS	PRIORITY
1	Two ML decoders that beat MWPM	Dr. Kim, QEC researchers	Essential
2	Stim data-generating pipeline	Dr. Kim	Essential
3	Trustworthy MWPM baseline	Dr. Kim, QEC researchers	Essential
4	All capstone deliverables	Team 22, Dr. Z	Essential
5	Python code and documentation	Dr. Kim, future students, Dr. Z	Important
6	Strong understanding of MWPM	Team 22	Important

Requirements Definition

Needs	Requirement	Units	Range	Ideal
Two ML models outperform MWPM	Two ML models outperform MWPM on the same test dataset that is statistically significant with a 95% confidence level.	%	>0%	TBD
Trustworthy MWPM	MWPM baseline produces similar logical error rates as published research	%	+/- 2%	+/- 1%
Simulated data matches physical computers	All datasets need to have realistic code distances and physical error rates	Code distance; error rate (%)	TBD	TBD
Written in Python	All code must be written and documented through Python	Programming Language	Python 3	Python 3
Strong understanding of MWPM	A documented handwritten analysis of MWPM with explanations of what it succeeds in and what it fails at	-	TBD	TBD
Completion of all Capstone Deliverables	Must complete all capstone deliverables on time	Seconds and days	<0s late	> 1 day early

References

[1] F. Battistel et al., “Real-time decoding for fault-tolerant quantum computing: Progress, challenges and outlook,” Nano Futures, vol. 7, no. 3, Art. no. 032003, Aug. 2023, doi: 10.1088/2399-1984/aceba6.

[2] J. Preskill, “Quantum computing in the NISQ era and beyond,” Quantum, vol. 2, p. 79, Aug. 2018, doi: 10.22331/q-2018-08-06-79.

[3] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, “Surface codes: Towards practical large-scale quantum computation,” Phys. Rev. A, vol. 86, no. 3, Art. no. 032324, Sep. 2012, doi: 10.1103/PhysRevA.86.032324.

[4] O. Higgott, “PyMatching: A Python package for decoding quantum codes with minimum-weight perfect matching,” ACM Trans. Quantum Comput., vol. 3, no. 3, pp. 1–16, 2022, doi: 10.1145/3505637.

[5] E. H. Chen et al., “Calibrated decoders for experimental quantum error correction,” Phys. Rev. Lett., vol. 128, no. 11, Art. no. 110504, Mar. 2022, doi: 10.1103/PhysRevLett.128.110504.

[6] J. Bausch et al., “Learning high-accuracy error decoding for quantum processors,” Nature, vol. 635, no. 8040, pp. 834–840, Nov. 2024, doi: 10.1038/s41586-024-08148-8.

[7] Google Quantum AI and Collaborators, “Quantum error correction below the surface code threshold,” Nature, vol. 638, no. 8052, pp. 920–926, Feb. 2025, doi: 10.1038/s41586-024-08449-y.

[8] C. Gidney, “Stim: A fast stabilizer circuit simulator,” Quantum, vol. 5, p. 497, Jul. 2021, doi: 10.22331/q-2021-07-06-497.

[9] IBM Quantum, “How IBM will build the world’s first large-scale, fault-tolerant quantum computer,” IBM Quantum Blog, Jun. 10, 2025. Accessed: Oct. 1, 2026. [Online]. Available: https://www.ibm.com/quantum/blog/large-scale-ftqc